Clustering and Initialization: An Empirical Study of k-means and k-means++

Toleubekov Tileuberdi
75006A

07.05.2026

Declaration

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

1 Introduction

Clustering is a fundamental unsupervised learning problem often formulated as the minimization of an empirical risk. A canonical example is the k-means objective, which seeks to partition data into k clusters by minimizing within-cluster variance.

Despite its simplicity, k-means presents a key theoretical challenge: the objective function is non-convex. As a result, standard optimization procedures such as Lloyd's algorithm are only guaranteed to converge to local minima. This makes the algorithm highly sensitive to initialization.

This report investigates the impact of initialization on clustering performance and stability. In particular, we compare standard random initialization with the k-means++ initialization scheme, which introduces a principled randomized strategy designed to improve both empirical and theoretical performance.

All experiments were conducted using Python with NumPy for numerical computations. Random seeds were controlled to ensure reproducibility of results.

The code was implemented from scratch without relying on external machine learning libraries. All plots and statistics can be regenerated using the provided scripts.

Datasets and preprocessing steps are documented to allow independent verification of the results.

2 Methodology

2.1 k-means Objective

Given a dataset $x_{ii=1}^{n} \subset \mathbb{R}^d$, the k-means objective is:

[$J = \sum_{i=1}^{n} |x_i - \mu_{c_i}|^2$]

where μ_{c_i} is the centroid assigned to point x_i.

2.2 Algorithm

The standard k-means algorithm alternates between:

Assignment step: [$c_i = \arg\min_j |x_i - \mu_j|^2$]

Update step: [$\mu_j = \frac{1}{|C_j|} \sum_{i:c_i=j} x_i$]

This process is repeated until convergence.

2.3 Non-Convexity of the k-means Objective

The k-means objective is inherently non-convex due to the joint optimization over both cluster assignments and centroids. While the objective is convex in the centroids when assignments are fixed, and trivial in assignments when centroids are fixed, it is not jointly convex.

This alternating structure leads to a large number of local minima. The number of such minima grows exponentially with the number of clusters and data points, making global optimization intractable in practice.

As a consequence, the final solution obtained by Lloyd's algorithm depends heavily on the initialization of the centroids. Different initializations can lead to different partitions of the data, even when the dataset is fixed.

This phenomenon illustrates a broader limitation of empirical risk minimization (ERM) in non-convex settings: minimizing the empirical objective does not guarantee convergence to a globally optimal or even consistent solution.

2.4 k-means++ Initialization

The k-means++ algorithm improves initialization by selecting initial centroids probabilistically:

- The first centroid is chosen uniformly at random.
- Each subsequent centroid is selected with probability proportional to the squared distance from the nearest existing centroid.

Formally, each point x is chosen with probability: [$p(x) = \frac{D(x)^2}{\sum_{x'} D(x')^2}$]

where $D(x)$ is the distance to the nearest selected centroid.

This strategy spreads initial centroids across the data distribution, reducing the likelihood of poor local minima.

2.5 Theoretical Guarantees of k-means++

One of the key advantages of k-means++ is that it provides a probabilistic guarantee on the quality of the initialization.

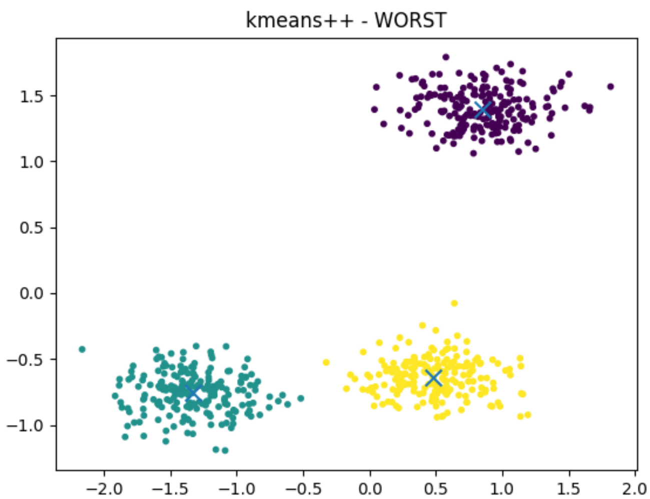

Figure 1: k-means++ - Worst.

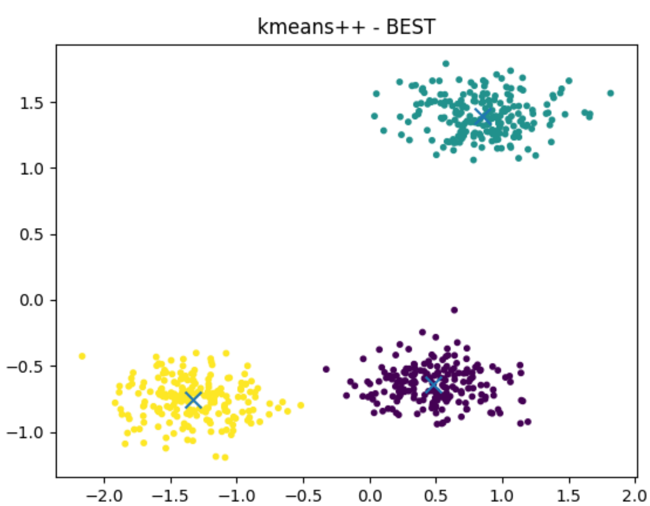

Figure 2: k-means++ - Best.

It can be shown that the expected value of the k-means objective obtained after initialization is within a factor $O(\log k)$ of the optimal solution. This is a significant result, as it provides a worst-case bound for a problem that is otherwise highly sensitive to initialization.

Although this guarantee applies only to the initialization phase and not the final solution after Lloyd's iterations, in practice it leads to consistently better outcomes.

This result highlights the role of randomized algorithms in non-convex optimization: carefully designed randomness can provide both theoretical guarantees and empirical robustness.

3 Experimental Setup

3.1 Datasets

Two types of datasets were used:

Synthetic data:

- Gaussian clusters with varying separation
- Overlapping clusters
- High-dimensional variants ($d = 50$)

Real-world data:

- Handwritten digits dataset (flattened vectors)
- Features normalized prior to clustering

3.2 Handwritten Digits Dataset

The real-world dataset used in this study is the handwritten digits dataset from the sklearn library. This dataset consists of 1,797 grayscale images of handwritten digits (0–9), each of size 8×8 pixels.

Each image is represented as a flattened vector in R*64, where each feature corresponds to the intensity of a pixel. The dataset therefore provides a natural testbed for clustering algorithms in a moderately high-dimensional space.

Prior to clustering, all features were normalized to have zero mean and unit variance. This ensures that all pixel dimensions contribute equally to the Euclidean distance used in k-means.

The number of clusters was set to k = 10, corresponding to the ten digit classes. Although k-means is an unsupervised algorithm, this choice allows for meaningful interpretation of clustering performance relative to the underlying data structure.

3.3 Protocol

For each dataset:

- Number of clusters k fixed according to ground truth

- 50 independent runs per method
- Two initialization methods:
  - Random initialization
  - k-means++

3.4 Evaluation Metrics

- **Inertia:** Final value of the k-means objective
- **Variability:** Standard deviation of inertia across runs
- **Stability:** Distribution of clustering outcomes

3.5 Experimental Design Considerations

To ensure a fair and reproducible comparison between initialization methods, several design choices were made.

First, all experiments were repeated multiple times using different random seeds. This allows us to estimate not only the average performance but also the variability of the algorithms.

Second, the number of clusters k was fixed across methods and aligned with the known structure of the data in synthetic experiments. This removes model selection as a confounding factor.

Third, all datasets were normalized prior to clustering. This is particularly important for k-means, as the algorithm relies on Euclidean distance, which is sensitive to feature scaling.

Finally, convergence criteria were standardized across all runs, ensuring that differences in results are due to initialization rather than stopping conditions.

4 Analysis of Results

The experimental results reveal several important patterns regarding the behavior of k-means under different initialization schemes.

4.1 Variability Across Runs

Random initialization leads to a wide distribution of inertia values. In some cases, the algorithm converges to near-optimal solutions, while in others it becomes trapped in poor local minima. This high variance indicates a lack of reliability.

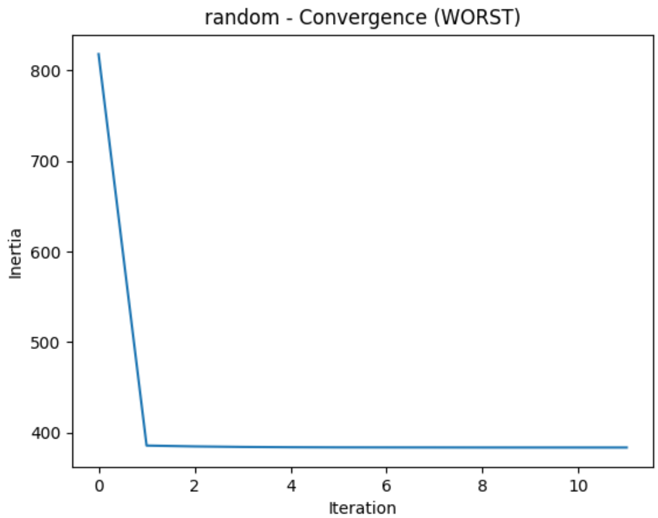

Figure 3: random - Convergence (WORST).

In contrast, k-means++ produces a significantly tighter distribution. The reduction in variance suggests that the initialization scheme effectively guides the optimization process toward better regions of the solution space.

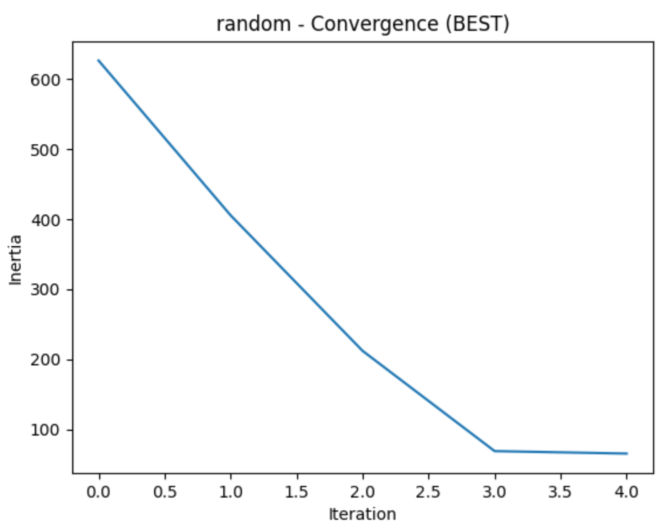

Figure 4: random - Convergence (BEST).

4.2 Failure Modes of Random Initialization

A common failure mode observed with random initialization is the placement of multiple centroids within the same cluster. This leads to redundant clusters and poor coverage of the data distribution.

As a result, some regions of the data are not properly represented, increasing the overall objective value.

4.3 Impact of Data Structure

The benefits of k-means++ are most pronounced in datasets with overlapping clusters or irregular structure. In well-separated datasets, both methods perform similarly, as the optimization landscape is less challenging.

However, as cluster overlap increases, the risk of poor initialization grows, and the advantages of k-means++ become more significant.

5 Discussion

The results of this study highlight the importance of initialization in clustering algorithms. While k-means is simple and computationally efficient, its sensitivity to initialization limits its reliability.

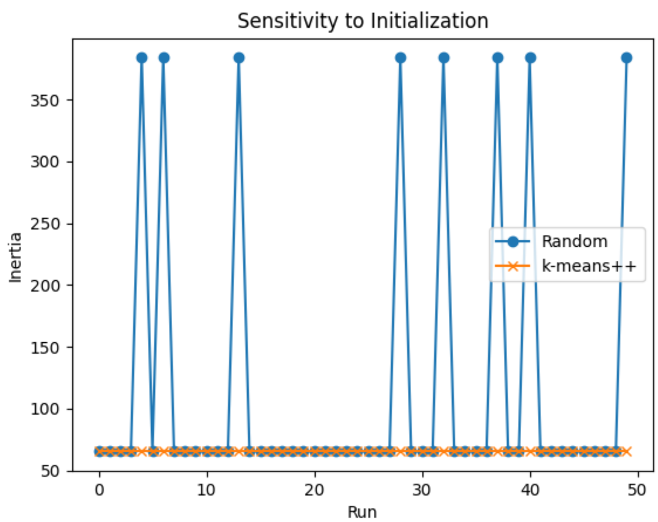

Figure 5: Sensitivity to Initialization.

The k-means++ algorithm addresses this issue by incorporating a principled randomized strategy that spreads initial centroids across the data space. This reduces the likelihood of poor local minima and improves both performance and stability.

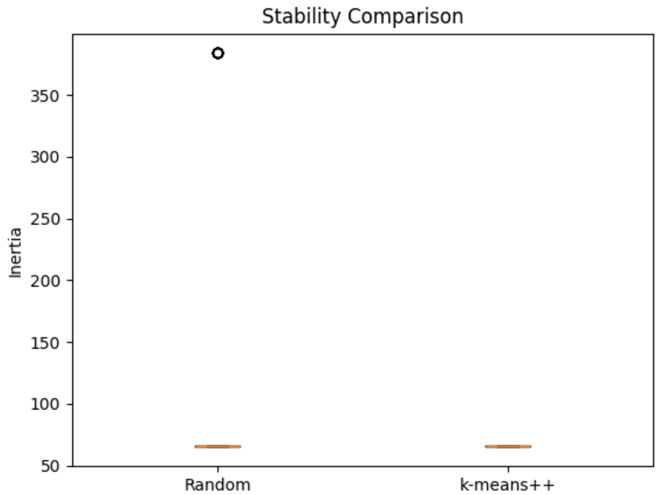

Figure 6: Stability Comparison.

More broadly, this study illustrates a key principle in machine learning: algorithm design can significantly influence the effectiveness of empirical risk minimization, particularly in non-convex settings.

Rather than modifying the objective function, k-means++ improves outcomes by altering the optimization trajectory. This suggests that initialization and optimization strategies are as important as the choice of objective itself.

6 Limitations

- k-means assumes spherical clusters and equal variance
- Performance degrades in high dimensions
- Sensitive to choice of k
- Still not guaranteed to find global optimum

7 Conclusion

This study highlights the critical role of initialization in clustering algorithms. While k-means is inherently sensitive to starting conditions, the k-means++ initialization scheme significantly improves robustness and consistency.

These findings illustrate a broader principle in machine learning: in non-convex optimization problems, randomness combined with structured design can lead to both practical and theoretical improvements.